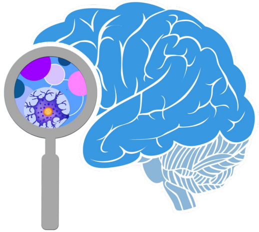
Human respiratory system

Lesson objectives:

- To describe the function of the major organs of the respiratory system such as trachea, lungs and alveoli.
- To list the organs in order that air passes through, as it moves into and out of the lungs.

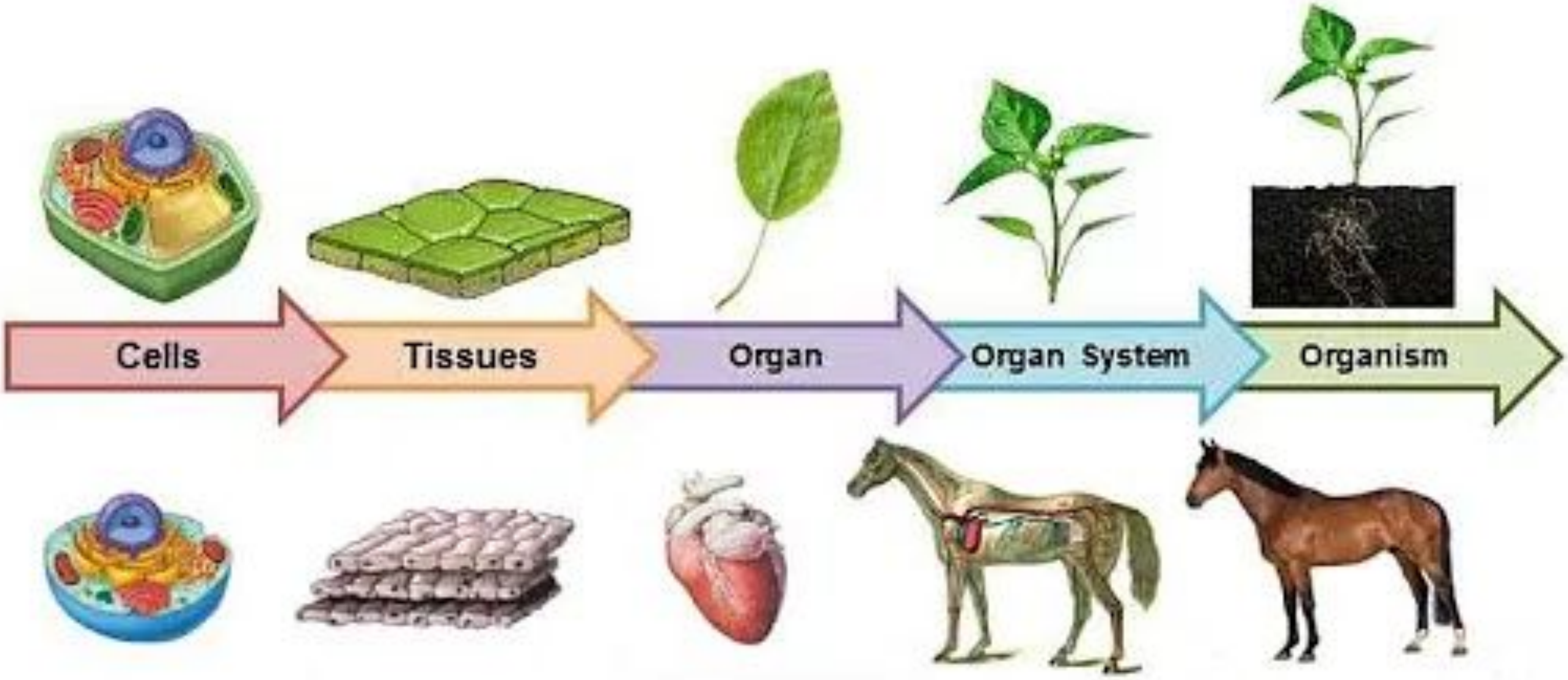


Thinking back to Year 7 …
What are the levels of organisation in an organism?

Success criteria

Lesson Title	Working towards	Working at	Working beyond
The human respiratory system	I can name the different parts of the human respiratory system, and identify them on a diagram.	I can list the organs that air passes through, as it moves into and out of the lungs.	I can explain how the adaptations of the parts of the human respiratory system help them perform their function.
Gas exchange	I can name the gases that make up air.	I can describe how gas exchange takes place in the alveoli.	I can explain why it is better to have lots of very small air sacs in the lungs, rather than a few big ones.
Breathing	I can use a model to show how breathing movements take place.	I can describe how the diaphragm and intercostal muscles cause breathing movements.	I can explain how these breathing movements make air move into and out of the lungs.
Aerobic Respiration	I can state that aerobic respiration happens inside mitochondria inside every living cell.	I can write the word equation for respiration.	I can explain the link between breathing and respiration.
Anaerobic Respiration	I can state that anaerobic respiration happens without oxygen.	I can describe the differences between aerobic and anaerobic respiration.	I can explain why anaerobic respiration produces lactic acid and how the body deals with it.
Blood	I can name the components of blood.	I can describe the appearance and function of the different components of blood.	I can explain how white blood cells help to protect us against pathogens.

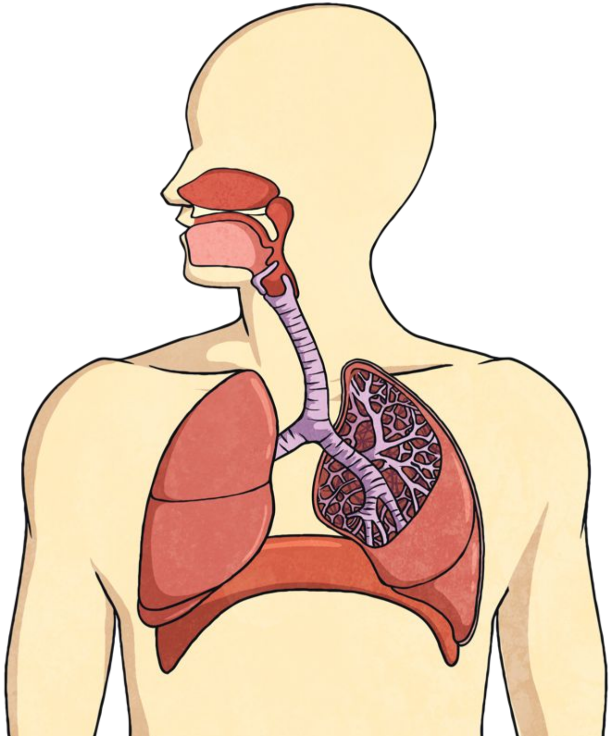
Levels of organisation



Respiratory system

The respiratory system contains:

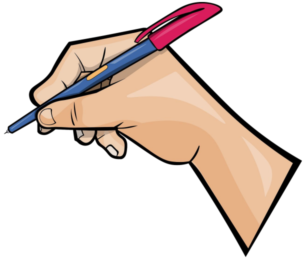
- Two lungs
- Tubes that lead from the entrance (mouth and nose) to the lungs
- Different structures in the chest that allow air to move in and out of the lungs



In groups of 4s, you are tasked to:

1. 

Sit In Circles

2. 

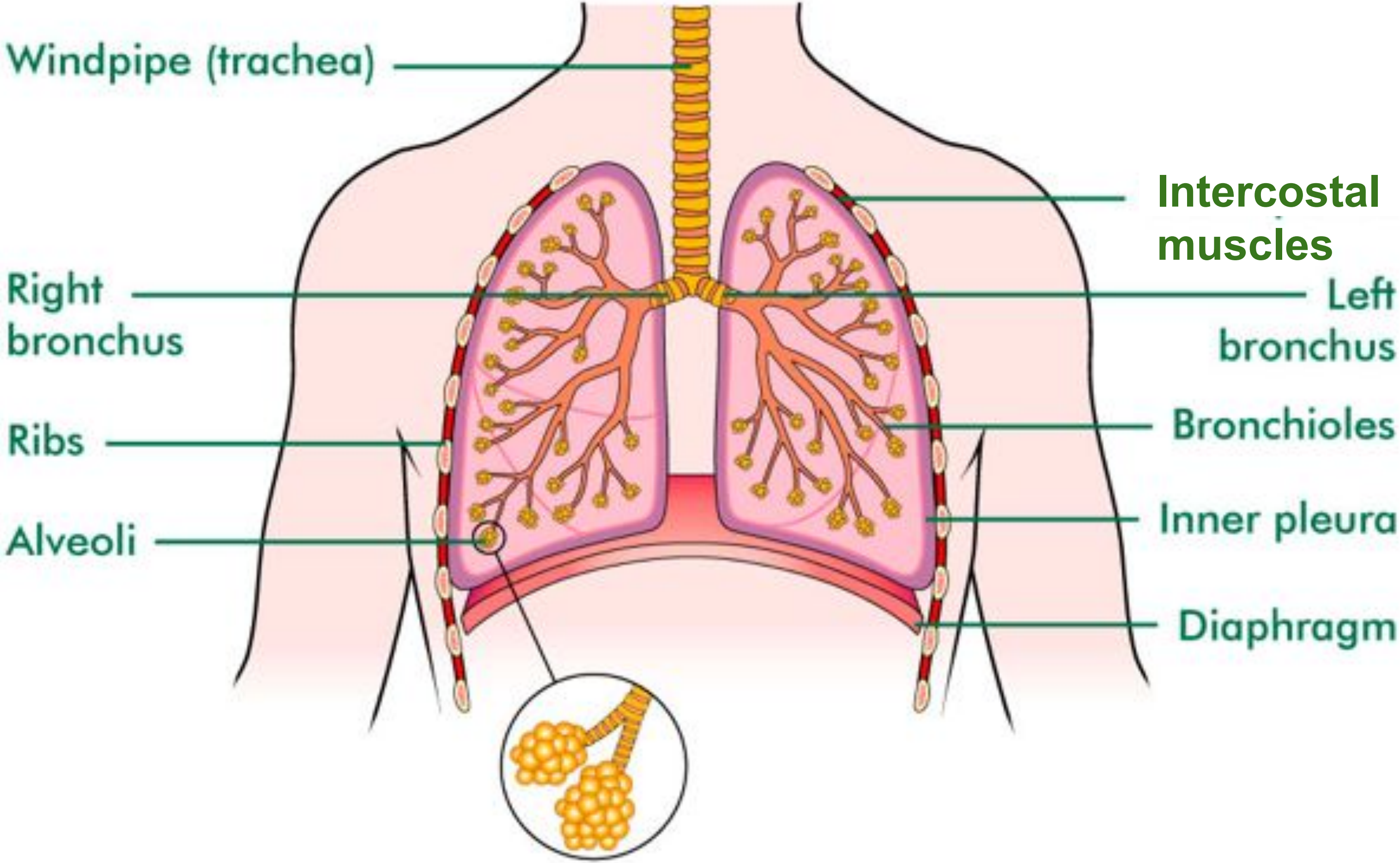
Have a pen in your hand

3. Each student in each group will have **30 seconds** to go outside of class to memorize what they can remember on the A3 sheet paper.

4. Once 30 seconds is up, you have to come back in to your group and write it down on your A3 sheet paper as much as you can remember

5. Then, the **next group member** have 30 seconds to go out and look at the picture and come back in.

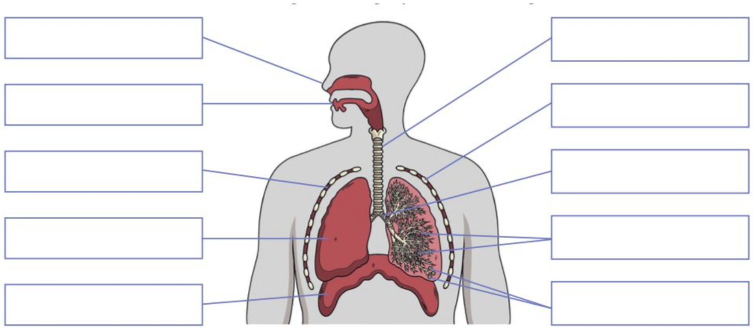
Respiratory system



Concept check!

Which structure is **not** part of the human respiratory system?

a) diaphragm
b) heart
c) lungs
d) trachea



Name of Part	Description

All: Add labels to the diagram

Most: Describe the function of each

Stretch: Your right lung is bigger than your left lung, can you think why?

Write your answer in your book (ask if you want a hint)

Textbook page 9 + 10

Respiratory system

Part of the gas exchange system	**Function**
Trachea	This is also called the windpipe. This tube runs from the mouth, down the throat towards the lungs. It is lined with rings of cartilage which keep it open at all times.
Bronchus	The trachea splits into a left and right bronchus (plural: bronchi), each leads to a lung.
Bronchiole	Each bronchus splits again and again into thousands of smaller tubes called bronchioles which take the air deeper into the lungs.
Alveoli	At the ends of bronchioles are tiny air sacs called alveoli. Here oxygen moves into the blood and carbon dioxide moves out.
Intercostal muscles	These muscles run between the ribs and form the chest wall. They contract and relax with the diaphragm when a person breathes.
Diaphragm	The diaphragm is a dome-shaped, flat sheet of muscle under the lungs. It contracts and relaxes with the intercostal muscles during breathing.

Concept check!

1. The lungs are an example of a tissue.
2. The trachea allows air to move from the nose and mouth into the lungs.
3. We have one lung.
4. The diaphragm is a muscle.
5. The lungs are big sacs of air.
6. The bronchi are smaller than the bronchioles.

Concept check!

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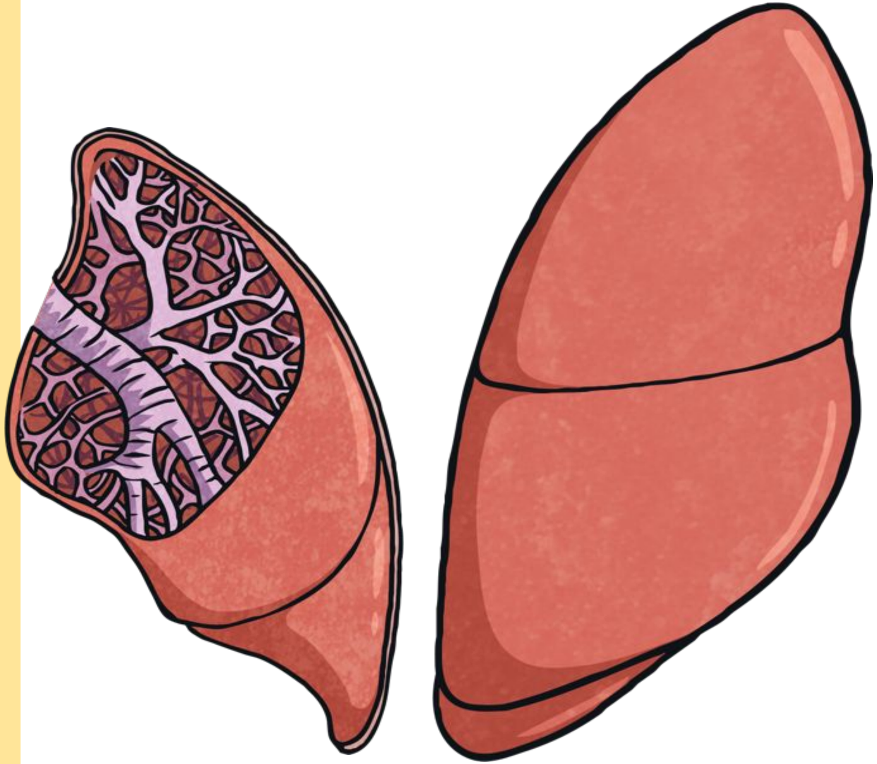
DID YOU KNOW?

Your lungs are located in your chest, inside your ribs. We have two lungs, but they are not the same size, like the way your eyes or ears are equal in size.

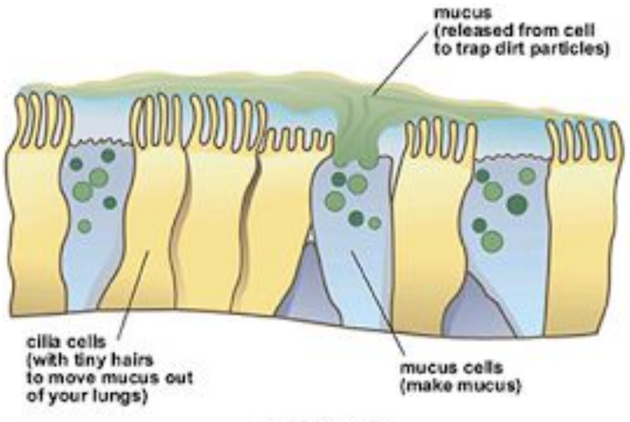
Your left lung is smaller than your right lung. This allows room for your heart, which is on the left side of the body.



Microscopic tissues in the respiratory system

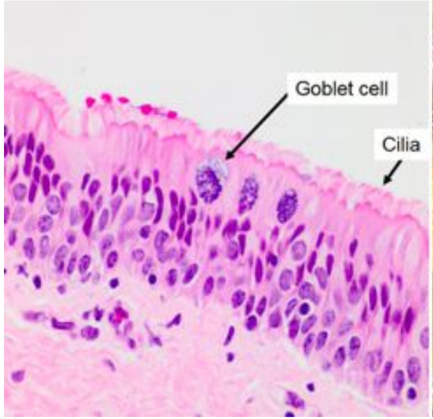
Goblet cells

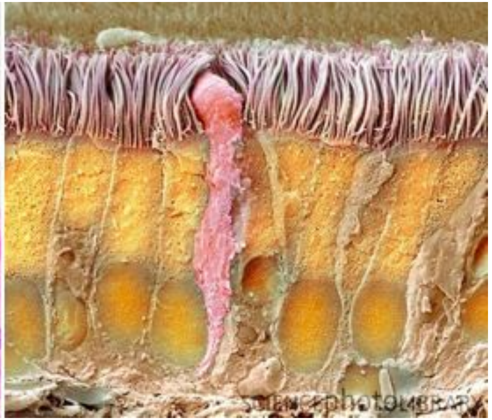
- Produce mucus (mucin)
- Mucus traps dust, dirt & pathogens



Ciliated cells

- Cilia beat rhythmically to move mucus up towards the mouth
- Mucus is then swallowed or coughed out





TO-DO **Describe** the journey of an oxygen molecule from the air into the lungs.

Sentence starter:

- **The oxygen molecule enters the mouth or nose.**
- **It then travels down the ________**

To 'describe', your answer should:

•Use **bullet points.**

•Include each step of the process in a **logical order.**

•Use **keywords**.

Stretch:

"The lungs are like big empty plastic bags inside our chest"

Is this statement correct or incorrect, **explain** your answer.

TO-DO

Describe the journey of an oxygen molecule from the air into the lungs.

- The oxygen molecule enters the **mouth or nose.**
- Then it travels down the **trachea.**
- The trachea branches into two tubes called **bronchi,** allowing oxygen to pass into the left and right lungs.
- The bronchi branch into **bronchioles,** which allows oxygen to reach the millions of tiny air sacs called **alveoli.**

We will learn this topic how oxygen can then move from the alveoli into our bloodstream.